

Auditing fast-mode molecular dynamics settings without fooling yourself

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Correction notice

Version 0.1.0 of this note (DOI 10.17605/OSF.IO/X4T8M) contains errors. Its main protein result, that **constraints = all-bonds** alone lets the villin protein reach a 6-7 fs timestep, is false: without hydrogen mass repartitioning (HMR), every all-bonds run at 5 fs or more crashed. The setting it recommended (6 fs, HMR factor 3, all-bonds) does pass the checks, but its protein validation tested a different HMR factor, its account of the h-bonds limit was wrong, its shell speedups were divided by a reference from a different batch, a units error meant the warmup was never excluded from the checks, and its Lyapunov figure was both miscalculated and confounded by a stochastic thermostat. This version corrects each error and re-validates the villin settings with three seeds each. Three rounds of independent AI review, of the released version and of two drafts of this correction, found these errors; the first draft itself contained a wrong recommendation, which the second round caught.

Abstract

Integration settings can make a molecular dynamics simulation several times faster, but the fastest setting that runs is not always correct. We present an open-source tool that searches the integration-settings space and validates every candidate against a 2 fs reference of the same system with four checks: conserved energy drift, temperature, the oxygen-oxygen radial distribution function, and density. We use it on pure water and on the villin HP-35 domain on a commodity CPU. Water reaches dt 5 fs (about 2.4x). For villin, h-bond constraints stop at dt 4 fs (about 1.9x), because **grompp** rejects heavy-atom bonds that are too fast for a larger step; hydrogen mass repartitioning cannot lift this limit, and at a mass factor of 4 it lowers it. All-bond constraints remove the limit but crash without HMR. With HMR factor 3 they reach dt 6 fs (about 2.7x), which passes all four checks in three seeds but shifts density, water structure and energy drift by small amounts that exceed the run-to-run noise. Tightening the LINCS constraint solver removes the drift bias but moves the water structure further, and that setting fails the checks in 6 of 9 comparisons. A spherical solvent shell reaches 4.6-8.8x but changes the model.

Statement of need

Molecular dynamics users raise the timestep, repartition hydrogen masses, switch on multiple time stepping, and shorten neighbour-list updates to gain speed. Each change can break the physics without an error message, and the difference between a thermal fluctuation and an integration error is often smaller than the run-to-run scatter. GROMACS refuses some settings at **grompp** time, but it does not tell the user whether the settings that did run are correct.

Systematic tests of physical validity exist: the **physical_validation** package (Merz and Shirts, PLOS ONE 2018) tests integrator convergence and ensemble validity. The tool described here is narrower. It searches over integration settings and compares every candidate with a same-system reference on four fixed checks, recording **grompp** refusals as refusals rather than forcing them with **-maxwarn**.

Method

Systems

We use three TIP3P water boxes of 2652, 8916 and 21087 atoms and the villin HP-35 domain in TIP3P, 14528 atoms (582 protein atoms, 4648 waters, 2 chloride ions, `amber99sb`). We also build a virtual-site variant of the protein with `pdb2gmx -vsite h -heavyh`. Runs use a development build of GROMACS (2027.0-dev) on an ARM CPU with four OpenMP threads and no GPU. No dispersion correction is applied, which lowers the absolute TIP3P density; reference and candidates share the choice. The thermostat is v-rescale with a strong coupling (`tau-t = 0.1 ps`).

The four checks

A candidate must satisfy all of the following against the 2 fs reference of the same system:

1. Conserved energy drift below 0.02 kJ/mol/ps per atom, from the GROMACS log line **Conserved energy drift** (whole run; in the re-validation the production-only drift agrees to within 0.0005).
2. Mean temperature within 1 K of the reference mean.
3. Maximum absolute deviation of the O-O radial distribution function below 0.02.
4. Mean density within 0.5 percent of the reference.

The first 2000 steps (4 ps at 2 fs, up to 14 ps at 7 fs) are meant to be excluded from checks 2-4; each run is that warmup plus the production time. Up to version 0.2.0 a units error excluded only the first frame. The audit, search and selfcheck results reported below carry that error, and the re-validation does not. A missing check is a failure. The RDF and density describe the water; for the protein we add a separate comparison of the radius of gyration, backbone RMSD and per-residue RMSF.

Noise floor

The tool's `selfcheck` mode runs three independent 2 fs replicas with different velocities and reports the largest pairwise difference for each check. At 200 ps:

system	temperature (K)	density (percent)	O-O RDF
water, 2652 atoms	0.931	0.172	0.0059
villin, 14528 atoms	0.722	0.103	0.0064

The tool reports these floors but does not use them in the verdict; a verdict is a fixed-threshold comparison. On small water the temperature floor nearly equals the 1 K limit, so a correct setting can fail by chance. A maximum over three samples is a rough estimate of the floor: the re-validation below, with the warmup correctly excluded, measures a villin density floor of up to 0.26 %, about 2.5 times the value in this table.

Timing

A run is timed only when the external machine load is below 2. This does not make timings comparable between batches: the same 2 fs villin reference has run at between 45.1 and 69.6 ns/day across batches. We report each speedup against a reference timed in the same batch. For the villin settings we timed every setting in five interleaved rounds and report the median ratio; those rounds ran under an external load of up to about 3.4, so the speedups are approximate.

Results

Water

All three water boxes reach dt 5 fs with no HMR, no MTS, `nstlist 10` and a buffer tolerance of 0.005, for 2.34x, 2.40x and 2.36x. MTS with PME every third step at dt 5 fs (a 15 fs slow-force interval) is unstable

on every box (drift about 0.5 against the 0.02 limit). A 20 ps probe on small water, checking drift and temperature only, passed every slow-force interval up to 12 fs and failed every interval of 15 fs or more. MTS was never faster than plain dt 5 fs.

Villin: where the limits are

h-bond constraints stop at dt 4 fs. `grompp` warns when a bond’s oscillation period is shorter than five timesteps. Without HMR it names the aspartate CG-OD1 bond (22 fs) at dt 5 fs. HMR moves mass from each hydrogen-bearing heavy atom to its hydrogens, which lightens that atom and speeds up its bonds to other heavy atoms. With HMR, `grompp` names tryptophan CG-CD1 (22 fs) at factors 2.5 and 3, lysine CE-NZ (21 fs) at 3.5, and leucine CG-CD1 at 4 (18 fs) and 4.5 (14 fs), where it rejects even dt 4 fs. The aspartate bond carries no hydrogens, so HMR cannot raise the limit above its 22 fs period. These are rule-of-thumb rejections; the runs were not attempted.

All-bond constraints need HMR. Constraining every bond removes the heavy-atom bond limit. Without HMR, every all-bonds run at dt 5 fs or more crashed. With HMR factors 2.5 to 4.5, all-bonds runs certified at dt 5, 6 and 7 fs, except factor 3 at dt 7 fs, which failed the RDF check (0.0206); every setting at dt 8 fs crashed. We have not identified what sets the dt 8 fs limit.

All-angles constraints fail. Every all-angles run stopped with too many LINCS warnings (1458 at dt 4 fs); coupled angle constraints do not converge in LINCS.

Virtual sites do not stack with all-bonds. A virtual-site rebuild reaches 2.74x at dt 6 fs, about the same as HMR plus all-bonds on the plain system. Its dt 7 fs candidate fails the RDF check (0.0299).

Villin: re-validation

We ran three seeds each of the 2 fs reference, plain dt 4 fs (h-bonds), and dt 6 fs with HMR factor 3 and all-bonds with default LINCS (order 4, one iteration) and with tight LINCS (order 8, two iterations), 200 ps each with the warmup excluded. We compared every candidate seed with every reference seed; the reference-to-reference differences give the noise floor (min / mean / max):

quantity	2 fs vs 2 fs	dt 4 fs, h-bonds	dt 6 fs, default LINCS	dt 6 fs, tight LINCS
density (%)	0.03 / 0.17 / 0.26	0.04 / 0.16 / 0.31	0.11 / 0.30 / 0.41	0.17 / 0.38 / 0.55
RDF max	0.002 / 0.004 /	0.008 / 0.011 /	0.010 / 0.012 /	0.017 / 0.021 /
deviation	0.006	0.016	0.014	0.025
Rg difference (nm)	0.004 / 0.006 / 0.008	0.000 / 0.004 / 0.008	0.001 / 0.008 / 0.021	0.000 / 0.004 / 0.009
RMSF correlation	0.83 / 0.86 / 0.90	0.67 / 0.83 / 0.95	0.81 / 0.89 / 0.98	0.72 / 0.86 / 0.97
energy drift	+0.0008	+0.0011	-0.0061	+0.0010
four checks passed		9 of 9	9 of 9	3 of 9

Plain dt 4 fs and dt 6 fs with default LINCS pass every comparison. Both shift the water structure above the noise floor and below the limit. dt 6 fs also raises the density by about 0.3 % in all three seeds and carries a systematic drift of -0.0061 from LINCS error, and one of its seeds has an Rg 1.7 % above the reference mean. Tight LINCS removes the drift bias but moves the water structure further (RDF deviation 0.017-0.025 against 0.010-0.014), and the setting fails the checks in 6 of 9 comparisons, with its mean RDF deviation (0.021) at the limit; its larger density shift is within the noise. We have not established why. Single runs at dt 7 fs pass 2 of 3 comparisons with default LINCS and none with tight LINCS. Every candidate group has RMSF correlations below the lowest reference pair (0.83): 0.67 for plain dt 4 fs, 0.81 for dt 6 fs with default LINCS and 0.72 with tight LINCS. Three runs per setting cannot resolve RMSF differences of this size, and the nine comparisons per setting share runs, so they are not independent.

In five interleaved timing rounds, plain dt 4 fs runs 1.93x (range 1.54-1.94x) and dt 6 fs with default LINCS 2.67x (range 2.13-2.70x), each relative to the same round’s 2 fs reference.

Version 0.1.0 reported one protein comparison for dt 6 fs all-bonds (RMSF correlation 0.90). That run used HMR factor 4, not factor 3, and there was no noise floor to interpret it.

The solvent shell: a model change

Removing bulk water and keeping only a solvent shell gives the largest numbers. A droplet of mobile water inside a box, with an outer water layer frozen or restrained, runs at dt 4 fs with all-bond constraints and no HMR. Against a periodic 2 fs h-bonds reference timed in the same batch (5 replicas of 0.3 ns, 55.8-69.5 ns/day, mean 64.8):

configuration	n	Rg (nm)	RMSF corr.	O-O order q	speedup
periodic, dt 2 fs	5	0.9614	1.000	0.557	1.0x
shell, PME, dt 4 fs	3	0.9627	0.969	0.403	4.6x (4.3-5.3x)
shell, reaction field 1.2 nm, dt 4 fs	5	0.9579	0.988	0.406	8.8x (8.2-10.3x)
shell, reaction field 1.5 nm, dt 4 fs	5	0.9583	0.968	0.405	6.9x (6.4-8.0x)

The ranges come from the spread of the reference replicas. The shell reproduces the protein observables, but the mobile water is interfacial (tetrahedral order about 0.40 against a bulk 0.557), and in the frozen variant the protein approaches the boundary. A position-restrained boundary moves the water toward bulk order (q 0.469 in a single run) but does not raise the dt 4 fs ceiling. Because the shell runs use all-bonds without HMR, and the periodic system in that state also crashes at dt 5 fs, the missing HMR is an untested candidate cause of that ceiling. The shell is valid for protein observables, not for bulk-solvent thermodynamics. Version 0.1.0 reported 7-13x for these configurations by dividing by a slower reference from a different batch.

Machine-learning probes

We tested GPU-free machine-learning routes to a general speedup:

route	outcome
learned next-position propagator	open: the divergence probe ran with a stochastic thermostat and does not measure a Lyapunov time; statistical accuracy was not tested
position-only force model	closed: R^2 0.81; the residual is PME plus the constraint force
learned MTS splitting	closed: the limit is a resonance, not a smooth bias
local water-patch proposal	closed: the Hastings penalty exceeds the gain
Boltzmann generator, Cartesian	closed: unphysical bond geometry; proposal energy about 1e18
Boltzmann generator, rigid body	closed: never learns excluded volume; acceptance 0/2000
one-site coarse-grained water	closed: iterative Boltzmann inversion diverges; the first RDF peak is 25 percent too low

Version 0.1.0 called the propagator route closed, citing a required accuracy of $1e-70$ for a 200 ps path; the same formula gives about $1e-696$. Both numbers rest on a divergence rate measured with a stochastic thermostat and a fresh random seed per run, which mixes thermostat noise into the growth, so the probe does not measure a Lyapunov time. And path accuracy is not what MD provides, so even a clean measurement would not close the route.

On this hardware a neural network is slower than optimized C per force evaluation, so a learned method can only win by reducing work or by generating samples, and the sample generators we tried failed on the sharp structure of a dense liquid.

Discussion

Every speed ingredient here is standard: HMR, all-bond constraints, virtual sites, reaction field and spherical boundaries. The contribution is an audit procedure and a record of what it found, including its own failures.

Those failures are the useful part. Version 0.1.0 passed its own checks and still drew a wrong conclusion: the write-up credited all-bond constraints without checking the all-bonds runs without HMR, which had crashed; the protein check ran on a different setting from the one recommended; single runs were compared without a noise floor; and a units error kept the warmup in three of the four checks. The first draft of this correction repeated the pattern: it recommended tighter LINCS settings on the strength of the drift alone, without running the four checks on them, and those settings fail. Replicated runs, a noise floor for every reported quantity, running every check on every recommended setting, and same-batch timing are what caught these errors. We recommend all four for any claim of this kind.

Constraining every bond is not a neutral choice: holonomic constraints change the protein’s configurational distribution, not only its dynamics. That does not explain the shifts we measured, which are in water density and structure; water is rigid in every run, and the search shows no extra density shift from all-bond constraints at dt 4 fs. The cause of the dt 6 fs shifts is not established.

Conclusion

For villin on a CPU, plain dt 4 fs with h-bonds (about 1.9x) passes the four checks with the smallest density and drift shifts. dt 6 fs with HMR factor 3, `constraints = all-bonds` and default LINCS settings (about 2.7x) also passes, with small measurable shifts in density (+0.3 %) and drift (-0.006); both settings shift the water structure by similar amounts. It suits questions those shifts do not affect. With tightened LINCS settings it fails the checks in 6 of 9 comparisons. Both are results for one small protein on one machine and need validation on other systems. A spherical shell reaches 4.6-8.8x but must be labelled a model change.

Data and code availability

The tool, the search harness, the validation and re-validation scripts, the input systems and the reduced results are at <https://github.com/dragon-str/fastmode-md> under the MIT license. Trajectories are not stored. The core tool, the re-validation, the timing and the `grompp` diagnosis run from the repository; the shell validation needs a shell system that is not included, and the machine-learning probes expect the original working-tree layout (see `README.md` and `m1/README.md`). Version 0.1.0 is registered on OSF at DOI 10.17605/OSF.IO/X4T8M.

Use of AI

AI coding agents (Claude models from Anthropic, and DeepSeek V4.1 Flash served by Fireworks AI) wrote most of the code, ran the simulations and drafted the text, working from the author’s written specification. AI reviewers found the errors that version 0.2.0 corrects. The author directed the work and is responsible for its content.

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